

# TM4C123 LED Blinky with UART Control

## README

A TM4C123GH6PM LaunchPad program that blinks the on-board RGB LED at a configurable rate and colour, shows that state on a 4-digit 7-segment display, and lets you control everything over UART instead of (or alongside) the two on-board switches.

### What it does

- Blinks the RGB LED (Port F) at one of 8 speeds and one of 8 colours.
- Drives a 4-digit 7-segment display: fixed 'S' label, current rate, current colour, and 'r' (running) / 'P' (paused).
- SW1 cycles the rate, SW2 cycles the colour, both together pause/resume.
- UART0 (115200-8-N-1) mirrors and controls the same state from a PC terminal.

### Hardware

TM4C123GH6PM LaunchPad; 7-segment display on Port A (digit select) and Port B (segments); on-board SW1/SW2 and RGB LED on Port F. The USB debug cable doubles as the UART0 link — no extra wiring.

### Using the UART Interface

Connect at **115200 baud, 8N1**, then type a command and press Enter:

| Command | Effect                                            |
|---------|---------------------------------------------------|
| RATE    | Advances the blink rate by one (wraps 7→0)        |
| COLOUR  | Advances the colour by one (wraps 7→0)            |
| PAUSE   | Pauses the lights (no effect if already paused)   |
| RESUME  | Resumes the lights (no effect if already running) |
| STATUS  | Reports the current rate, colour, and state       |

Commands are **case-sensitive** — type them in capitals exactly as shown.

### Notes

- Switches and UART commands drive the same state, so either input is reflected immediately.
- PAUSE/RESUME sent when already in that state prints a short message instead of changing anything.
- The STATUS reply prints the literal word "status" with no line break right before the status line — a known quirk of the current handler, not a hardware fault.